



COMP – 304

Summer 2026



Introduction to Android Platform and Android Application Development

Objectives:

- ☐ Examine Android Platform.
- ☐ Discuss the differences between leading mobile operating systems.
- ☐ Explain Android Development Environment and Android Application Architecture.
- ☐ Write a simple Android application using Android Studio.



What is Android

- ❑ Android is an **open source software stack** for a wide range of mobile devices and a corresponding **open source project** led by Google.
- ❑ It also includes **a rich set of API libraries** that make it possible to write applications.
- ❑ **Andy Rubin** has been credited as the father of the Android platform.
 - His company, Android Inc., was acquired by Google in 2005.
 - Android was announced in 2007.
 - The first Android mobile handset, the T-Mobile G1, was released in the United States in **October 2008**.
- ❑ Android is an **open source** platform
 - No need to pay royalties or license fees to develop for the platform.



Android Open Source Project: A Triad Ecosystem

- ❑ Google stewards the Android Open Source Project, offering a suite of resources for developers including **Android SDK, documentation, tools, and community forums**.
- ❑ **News and Innovations:** Major updates and news about Android often come from Google, yet the vibrant ecosystem also includes contributions from various manufacturers and the developer community at large.
- ❑ **Android Ecosystem:** The Android ecosystem encompasses:
 1. A **free, versatile open-source** operating system not only for mobile but also for a range of embedded devices.
 2. A **robust open-source development platform** equipped with tools like Android Studio and SDK to craft applications.
 3. A **diverse spectrum of devices** powered by Android OS, supporting a multitude of applications tailored for different user experiences.



Android Versions

Name	Internal codename ^[11]	Version number(s)	API level	Release date	Latest security patch date ^[16]	Latest Google Play Services version (release date)
Android 1.0	—	1.0	1	September 23, 2008	—	—
Android 1.1	Petit Four	1.1	2	February 9, 2009		
Android Cupcake	Cupcake	1.5	3	April 27, 2009		
Android Donut	Donut	1.6	4	September 15, 2009		
Android Eclair	Eclair	2.0	5	October 27, 2009		
		2.0.1	6	December 3, 2009		
		2.1	7	January 11, 2010 ^[17]		
Android Froyo	Froyo	2.2 – 2.2.3	8	May 20, 2010		3.2.25 (October 2014)
Android Gingerbread	Gingerbread	2.3 – 2.3.2	9	December 6, 2010		10.0.84 (November 2016)
		2.3.3 – 2.3.7	10	February 9, 2011		
		3.0	11	February 22, 2011		
Android Honeycomb	Honeycomb	3.1	12	May 10, 2011		14.8.49 (February 2019)
		3.2 – 3.2.6	13	July 15, 2011		
		4.0 – 4.0.2	14	October 18, 2011		
Android Ice Cream Sandwich	Ice Cream Sandwich	4.0.3 – 4.0.4	15	December 16, 2011		21.33.56 (September 2021)
		4.1 – 4.1.2	16	July 9, 2012		
Android Jelly Bean	Jelly Bean	4.2 – 4.2.2	17	November 13, 2012		
		4.3 – 4.3.1	18	July 24, 2013		
Android KitKat	Key Lime Pie	4.4 – 4.4.4	19	October 31, 2013	October 2017	23.30.13 (August 2023)
		4.4W – 4.4V.2	20	June 25, 2014	?	
Android Lollipop	Lemon Meringue Pie	5.0 – 5.0.2	21	November 4, 2014 ^[18]	November 2017	24.13.18 (April 2024)
		5.1 – 5.1.1	22	March 2, 2015 ^[19]	March 2018	
Android Marshmallow	Macadamia Nut Cookie	6.0 – 6.0.1	23	October 2, 2015 ^[20]	August 2018	
Android Nougat	New York Cheesecake	7.0	24	August 22, 2016	August 2019	
		7.1 – 7.1.2	25	October 4, 2016	October 2019	
Android Oreo	Oatmeal Cookie	8.0	26	August 21, 2017	January 2021	
		8.1	27	December 5, 2017	October 2021	
Android Pie	Pistachio Ice Cream ^[21]	9	28	August 6, 2018	January 2022	
Android 10	Quince Tart ^[22]	10	29	September 3, 2019	February 2023	
Android 11	Red Velvet Cake ^[23]	11	30	September 8, 2020	February 2024	
Android 12	Snow Cone	12	31	October 4, 2021	April 2024	
Android 12L	Snow Cone v2	12.1 ^[24]	32	March 7, 2022		
Android 13	Tiramisu	13	33	August 15, 2022		
Android 14	Upside Down Cake ^[25]	14	34	October 4, 2023		
Android 15	Vanilla Ice Cream ^[26]	15 Beta 1 ^[27]	35	April 11, 2024 ^[27]	April 2024 ^[27]	24.11.13 (March 2024) ^[27]
Legend: Old version Older version, still maintained Latest version Latest preview version						



Android Versions

- ❑ **Android's dessert-themed journey** began with **Cupcake** and continued alphabetically through notable releases like **Donut**, **Éclair**, **Froyo**, **Gingerbread**, **Honeycomb**, **Ice Cream Sandwich**, **Jelly Bean**, **KitKat**, **Lollipop**, **Marshmallow**, **Nougat**, **Oreo**, and **Android 9 Pie** in August 2018.
- ❑ The convention changed with **Android 10**, which was unveiled on June 5, 2019, adopting a simpler numerical naming system.
- ❑ This was followed by Android 11, launched on September 8, 2020, continuing the numeric trend for version names.
- ❑ Android's **current version is 15**, April 11, 2024.





Applications and downloads

Year	Month	Applications available	Downloads to date
2009	March	2,300 ^[109]	
	December	16,000 ^[110]	
2010	March	30,000 ^[111]	
	April	38,000 ^[112]	
	July	70,000 ^[113]	
	September	80,000 ^[114]	
	October	100,000 ^[115]	
2011	April		3 billion ^[116]
	May	200,000 ^[117]	4,5 billion ^[117]
	July	250,000 ^[118]	6 billion ^[118]
	October	500,000 ^{[119][120]}	
	December		10 billion ^[121]
2012	April		15 billion ^[122]
	June	600,000 ^[123]	20 billion ^[123]
	September	675,000 ^[124]	25 billion ^[124]
	October	700,000 ^[125]	
2013	May		48 billion ^[126]
	July	1 million ^[127]	50 billion ^[127]
2016			82 billion ^[128]
2017	February	2.7 million ^[1]	



Android Market Share Statistics

- ❑ Android has a share of **71.77%**, while iOS accounted for 27.6% of the mobile OS market.
 - There are more than 2.5 billion android active users in more than 190 countries.
 - A total of 1.57 billion Android devices sold in 2022, up from 0.98 billion as of 2021.
- ❑ Android accounted for 44.86% of the mobile OS market, while iOS accounted for 54.74% as of January 2023.
- ❑ Android sales are rising in the US, Europe, Japan, India, and China.
- ❑ Samsung's market share in the US was 28.57% in 2023, which is raised from 26.45% in 2022.
- ❑ Android's share of all known devices was 70.97%.

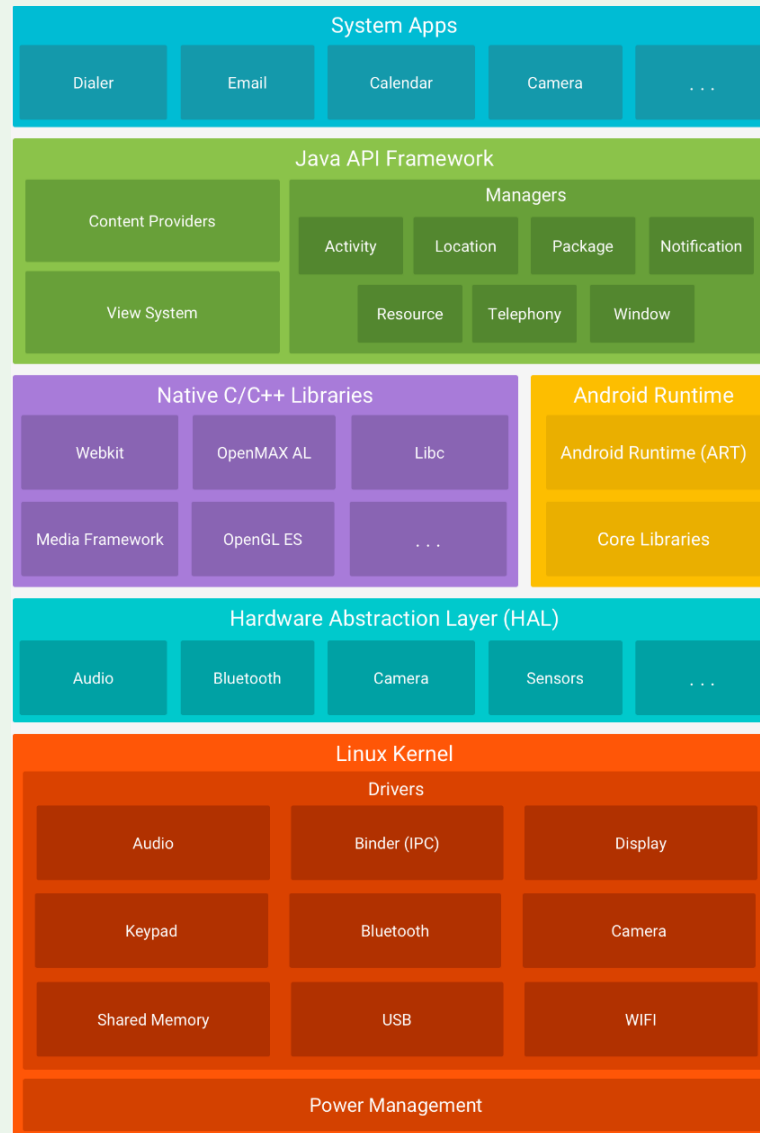


Android platform

- ❑ Android is an **operating system and a software platform** upon which applications are developed
- ❑ The Android platform is best described as a *stack because it is a collection of* components, including:
 - **Linux** kernel-based operating system
 - **Hardware Abstraction Layer (HAL)**
 - **Android Runtime (ART)**
 - **Native C/C++ Libraries**
 - **Java API Framework**
 - **Systems Apps**
- ❑ See the picture in next page for details



Android platform – Software Stack





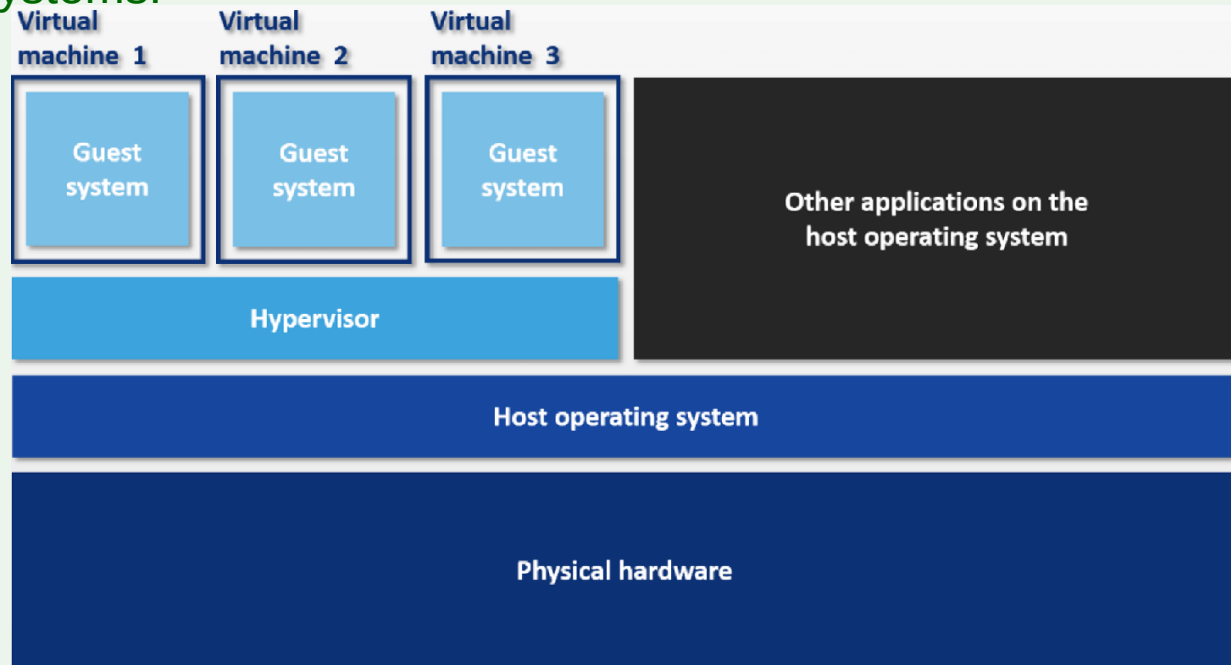
Android platform – Software Stack

- ❑ **Linux kernel** - it contains all the **essential hardware drivers** like camera, keypad, display etc.
- ❑ **Android HAL** - allows the Android libraries to communicate with the hardware specific device drivers.
- ❑ **Android Native C/C++ Libraries** – WebKit, OpenGL, Media, etc.
- ❑ **Android RunTime (ART)** – core libraries and runtime environment.
- ❑ **Java API Framework** - higher-level services to applications in the form of Java classes.
- ❑ **Applications** - your application to be installed on this layer only.



Android and Virtual Machines

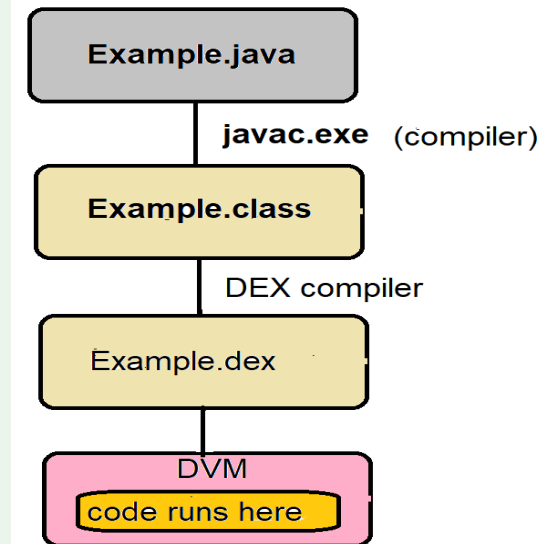
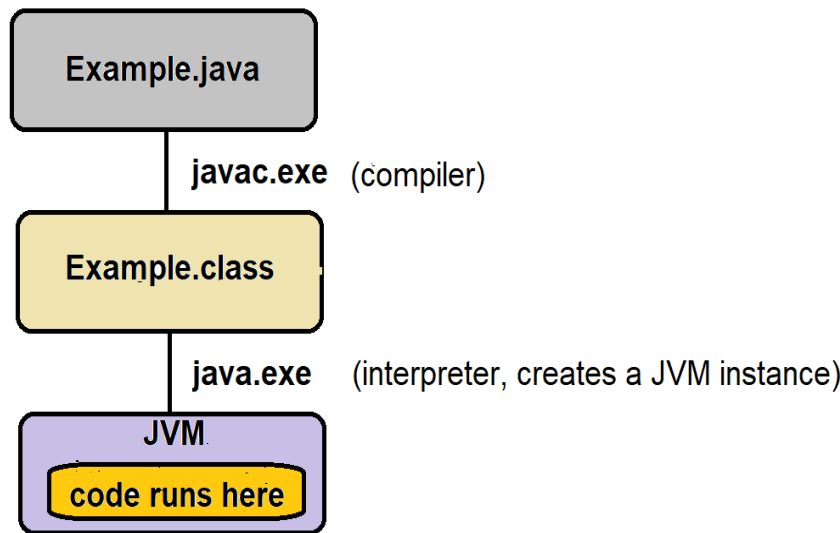
- ❑ **VMs emulate computer systems**, providing the functionality of physical hardware within a virtual environment.
- ❑ They are **built upon computer architectures**, allowing them to replicate the computational processes of physical computers..
- ❑ System virtual machines (or full virtualization VMs) provide a substitute for a real machine.
 - They provide functionality needed to execute entire operating systems.





Android and Virtual Machines

- ❑ **Process virtual machines (like JVM)** allow programs to execute in a platform-independent environment.
 - Implementing the same JVM on multi **platforms** - applications that it executes can be used on all those platforms.

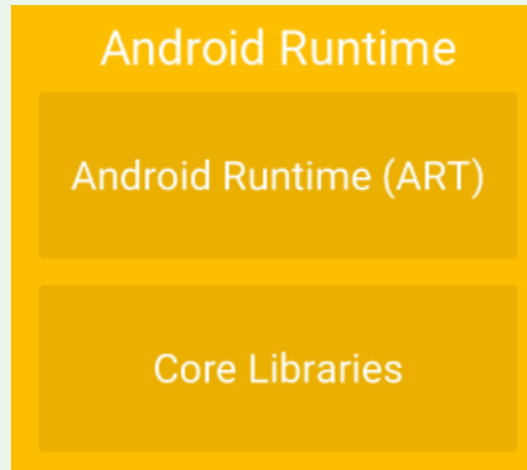


- ❑ Android's **Dalvik VM (DVM)** was created specifically for mobile device environment.
- ❑ A (Dalvik executable file) **DEX file** is a **translated from bytecodes specially for Android** and is optimized for minimal memory footprint.



Android Runtime

- ❑ ART replaced Dalvik starting with Android 5 (Lollipop).
- ❑ ART **runs multiple virtual machines on low-memory devices by executing (Dalvik executable files) DEX files.**



- ❑ **.dex** executables are created by transforming Java or Kotlin language compiled classes using SDK tools.
- ❑ Some of the major features of ART include the following:
 - **Ahead-of-time (AOT)** and **just-in-time (JIT)** compilation
 - Optimized **garbage collection (GC)**
 - Better **debugging** support



JIT versus AOT

- ❑ **Dalvik Just In Time (JIT) compiler**, every time that the app is running:
 - **dynamically translates a portion of the Dalvik bytecode into machine code and caches it**
 - uses less physical space on the device.
 - Then it **takes the next portion** and so on
- ❑ **ART Ahead Of Time (AOT) compiler**, when the app is installed on the device:
 - **statically translates the DEX bytecode into machine code** and stores in the device's storage.
 - the code executes **much faster**
 - **less battery drain** because uses native execution, uses CPU less than JIT
- ❑ The JIT compiler **complements** AOT compiler, saves storage space, and speeds application and system updates.



Apple's iOS vs Android

- ❑ The **iOS** is the operating system created by Apple Inc. for mobile devices.
- ❑ The basic functions in iOS and Android are the same.
- ❑ iOS is a **closed system** whereas Android is more open.
 - Users have barely any system permissions in iOS but in Android, users can customize their phones easily.
- ❑ Android software is available for many manufacturers such as Samsung, LG etc. and this may lead to some quality problems in the cheaper phones.



Application Fundamentals

- ❑ Android apps can be written using Java, Kotlin, and C++ languages.
- ❑ The Android SDK tools **compile your code** along with any data and resource files **into an APK**, an **Android package**, which is an archive file with an **.apk** suffix.
- ❑ One APK file contains all the contents of an Android app and is **the file that Android-powered devices use to install the app**.
- ❑ By default, **every app runs in its own Linux process**.
- ❑ Each process has its own virtual machine (VM), so an **app's code runs in isolation from other apps**.



Development Environment

- ❑ You need Android SDK and a Java Development Kit (JDK).
- ❑ **Android Studio** is the official Android IDE, based on IntelliJ IDEA.
- ❑ Install the latest Android Studio from this link:
<https://developer.android.com/studio>

```
1 package com.example.greetingcard
2
3 import android.os.Bundle
4 import androidx.activity.ComponentActivity
5 import androidx.activity.compose.setContent
6 import androidx.compose.foundation.layout.fillMaxSize
7 import androidx.compose.material3.MaterialTheme
8 import androidx.compose.material3.Surface
9 import androidx.compose.material3.Text
10 import androidx.compose.runtime.Composable
11 import androidx.compose.ui.Modifier
12 import androidx.compose.ui.tooling.preview.Preview
13 import com.example.greetingcard.ui.theme.GreetingCardTheme
14
15 class MainActivity : AppCompatActivity() {
16     override fun onCreate(savedInstanceState: Bundle?) {
17         super.onCreate(savedInstanceState)
18         setContent {
19             GreetingCardTheme {
20                 // A surface container using the 'background' color from the theme
21                 Surface(modifier = Modifier.fillMaxSize(), color = MaterialTheme.colorScheme.background) {
22                     Greeting(name = "Android")
23                 }
24             }
25         }
26     }
27 }
28
29 @Composable
30 fun Greeting(name: String, modifier: Modifier = Modifier) {
31     Text(
32         text = "Hello $name!",
33         modifier = modifier
34     )
35 }
```



Android Emulator

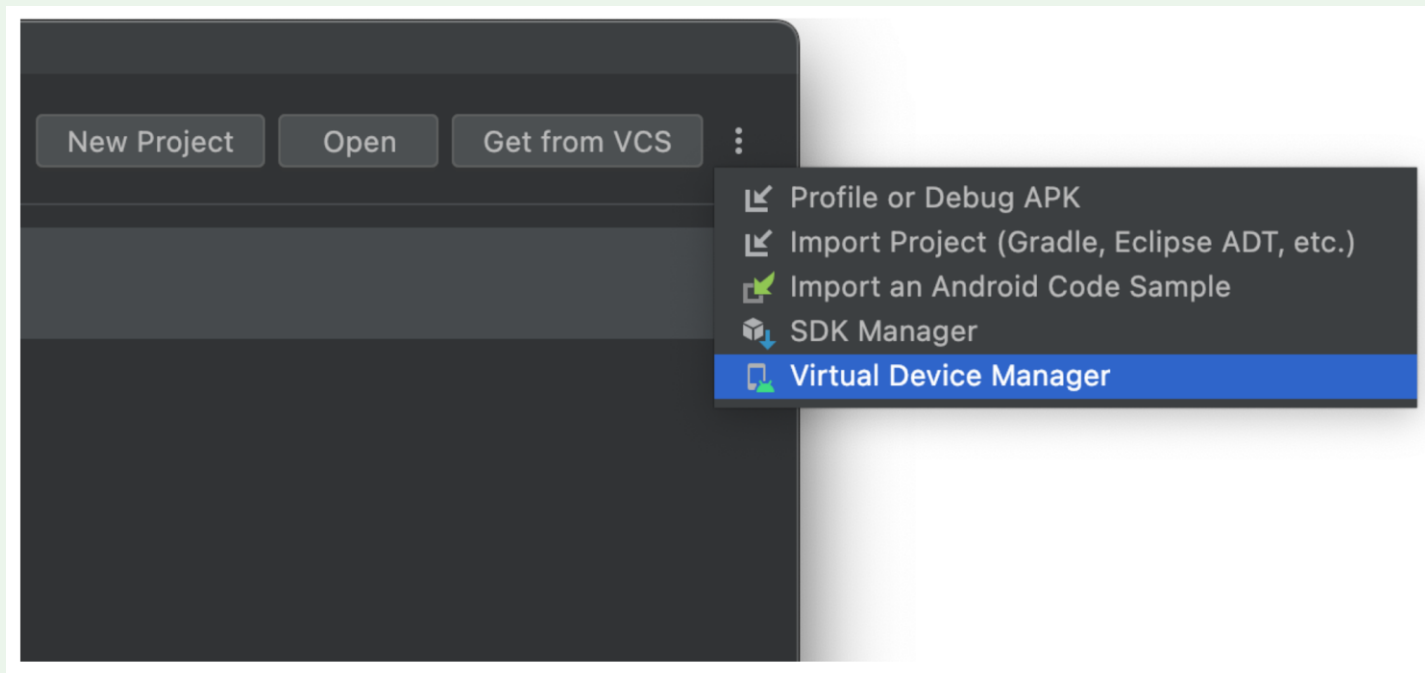
- ☐ The Android **emulator**, is one of the most important tools provided with the Android SDK.
- ☐ You will use this tool frequently when designing and developing Android applications.
- ☐ The emulator runs on your computer and behaves much as a **mobile device** would.
- ☐ You can **load** Android applications into the emulator, **test**, and **debug** them.





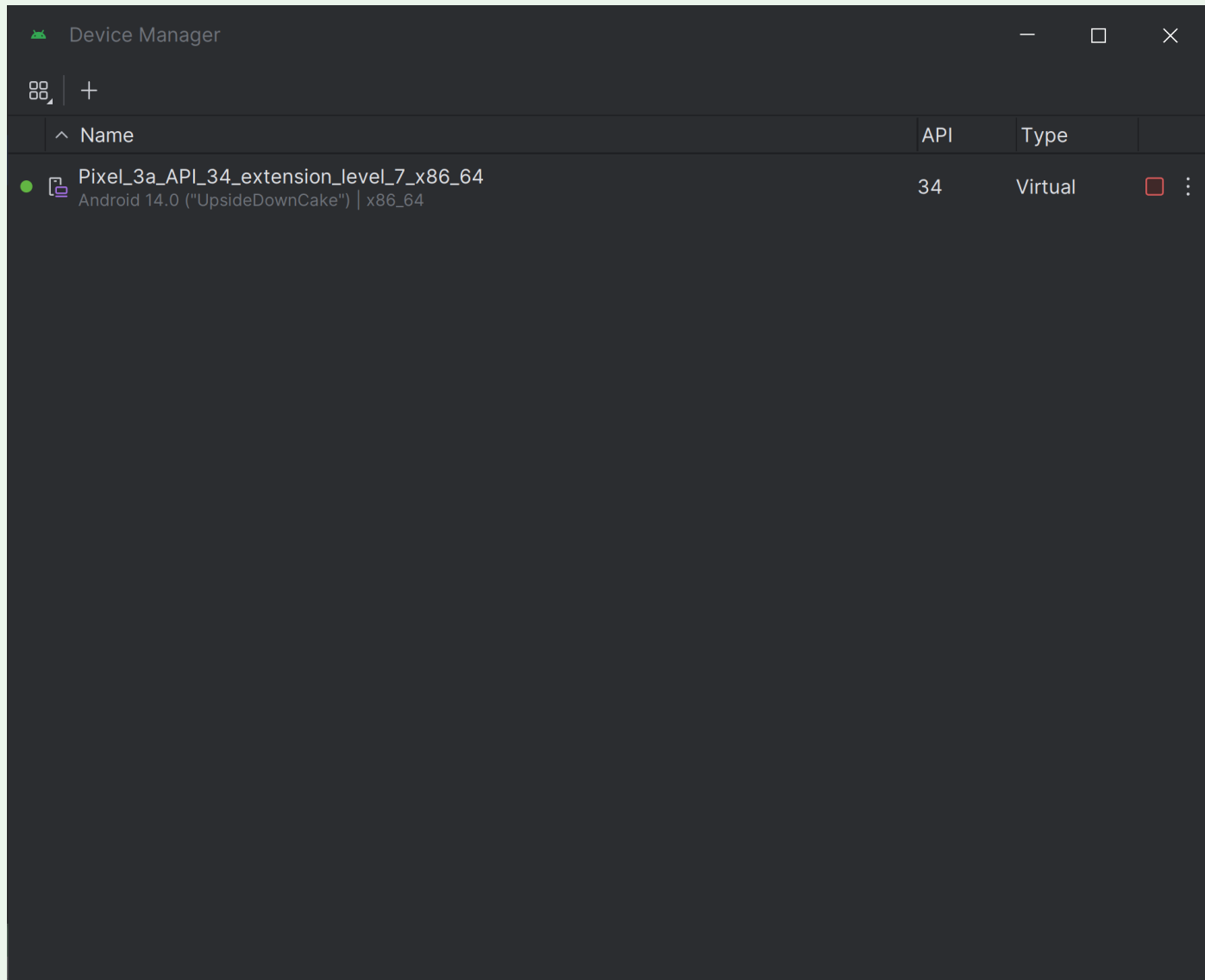
AVD Manager

- ❑ The AVD Manager is a tool you can use to create and manage Android virtual devices (AVDs), which define device configurations for the Android Emulator.
- ❑ To launch the AVD Manager: From the Android Studio Welcome screen, select More Actions > Virtual Device Manager





AVD Manager





Creating an AVD

- ❑ You can create as many AVDs as you would like to use with the Android Emulator.
- ❑ To effectively test your app, you should create an AVD that models each device type for which you have designed your app to support.
 - For instance, you should create an AVD for each API level equal to and higher than the minimum version you've specified in your manifest `<uses-sdk>` tag.
- ❑ To create an AVD based on an existing device definition:
 - Click the + icon on the Device Manager screen.
 - In the Select Hardware window, select a device configuration, such as Google Pixel 7a, then click Next.



Creating an AVD

Virtual Device Configuration

System Image

Select a system image

Recommendedx86 ImagesOther Images

Release Name	API Level	ABI	Target
VanillaIceCream	VanillaIceCream	x86_64	Android API VanillaIceC...
UpsideDownCa...	UpsideDownCak	x86_64	Android API UpsideDov
TiramisuPrivac...	TiramisuPrivacy	x86_64	Android 14.0 (Google P
UpsideDownCake	34	x86_64	Android 14.0 (Google F
Tiramisu	33	x86_64	Android 13.0 (Google P
Sv2	32	x86_64	Android 12L (Google PI
S	31	x86_64	Android 12.0 (Google P
R	30	x86	Android 11.0 (Google P
Q	29	x86	Android 10.0 (Google P

UpsideDownCake

API Level

34

Type

Google Play

Android

14.0

Google Inc.

System Image

x86_64

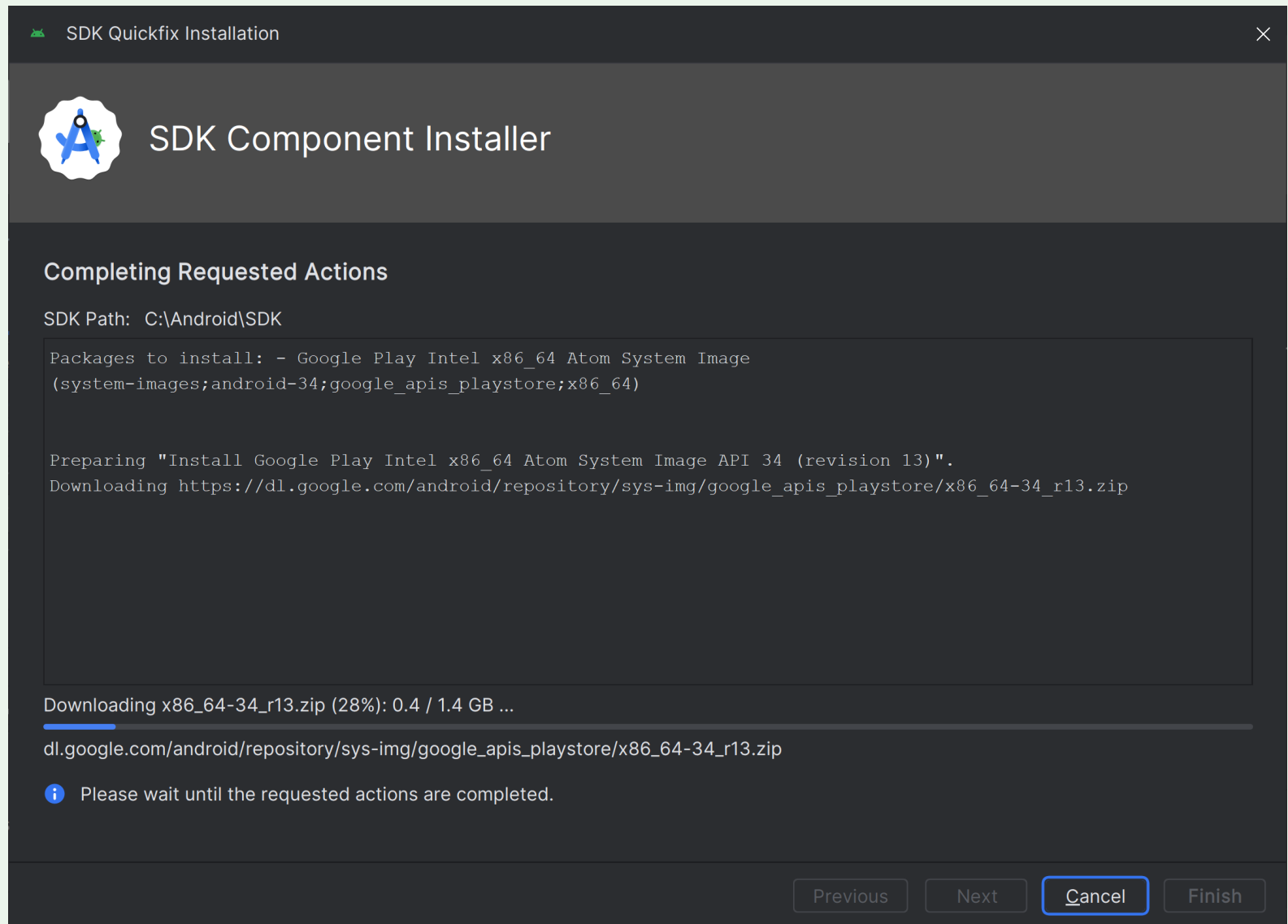
We recommend these Google Play images because this device is compatible with Google Play.

A system image must be selected to continue.

PreviousNextCancelFinish



Installing a System Image





Installing a System Image

Virtual Device Configuration

System Image

Select a system image

Recommended x86 Images Other Images

Release Name	API Level	ABI	Target
VanillaIceCream	VanillaIceCream	x86_64	Android API VanillaIceC...
UpsideDownCake	UpsideDownCake	x86_64	Android API UpsideDov
TiramisuPrivacyS...	TiramisuPrivacyS...	x86_64	Android 14.0 (Google P
UpsideDownCake	34	x86_64	Android 14.0 (Google P
Tiramisu	33	x86_64	Android 13.0 (Google P
Sv2	32	x86_64	Android 12L (Google P
S	31	x86_64	Android 12.0 (Google P
R	30	x86	Android 11.0 (Google P
Q	29	x86	Android 10.0 (Google P

UpsideDownCake

API Level
34

Type
Google Play

Android
14.0

Google Inc.

System Image
x86_64

We recommend these Google Play images because this device is compatible with Google Play.

Previous Next Cancel Finish



Verifying AVD Configuration

 Android Virtual Device (AVD)

Verify Configuration

AVD Name

Pixel 7a API 34

 Pixel 7a

6.1 1080x2400 420dpi

Change...

 UpsideDownCake

Android 14.0 x86_64

Change...

Startup orientation

 Portrait

 Landscape

Emulated Performance

Graphics: Automatic

Show Advanced Settings

AVD Name

The name of this AVD.

?

Previous

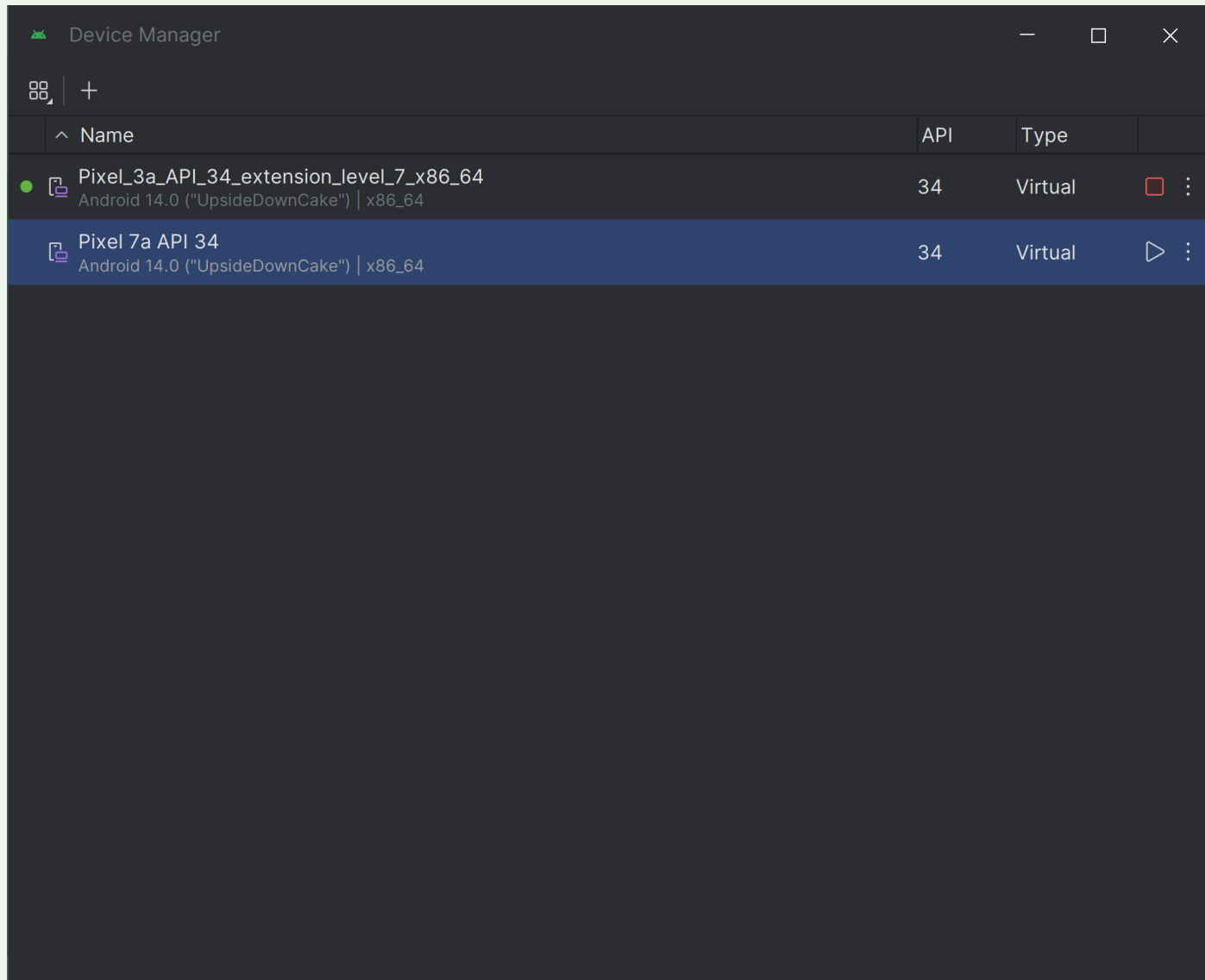
Next

Cancel

Finish

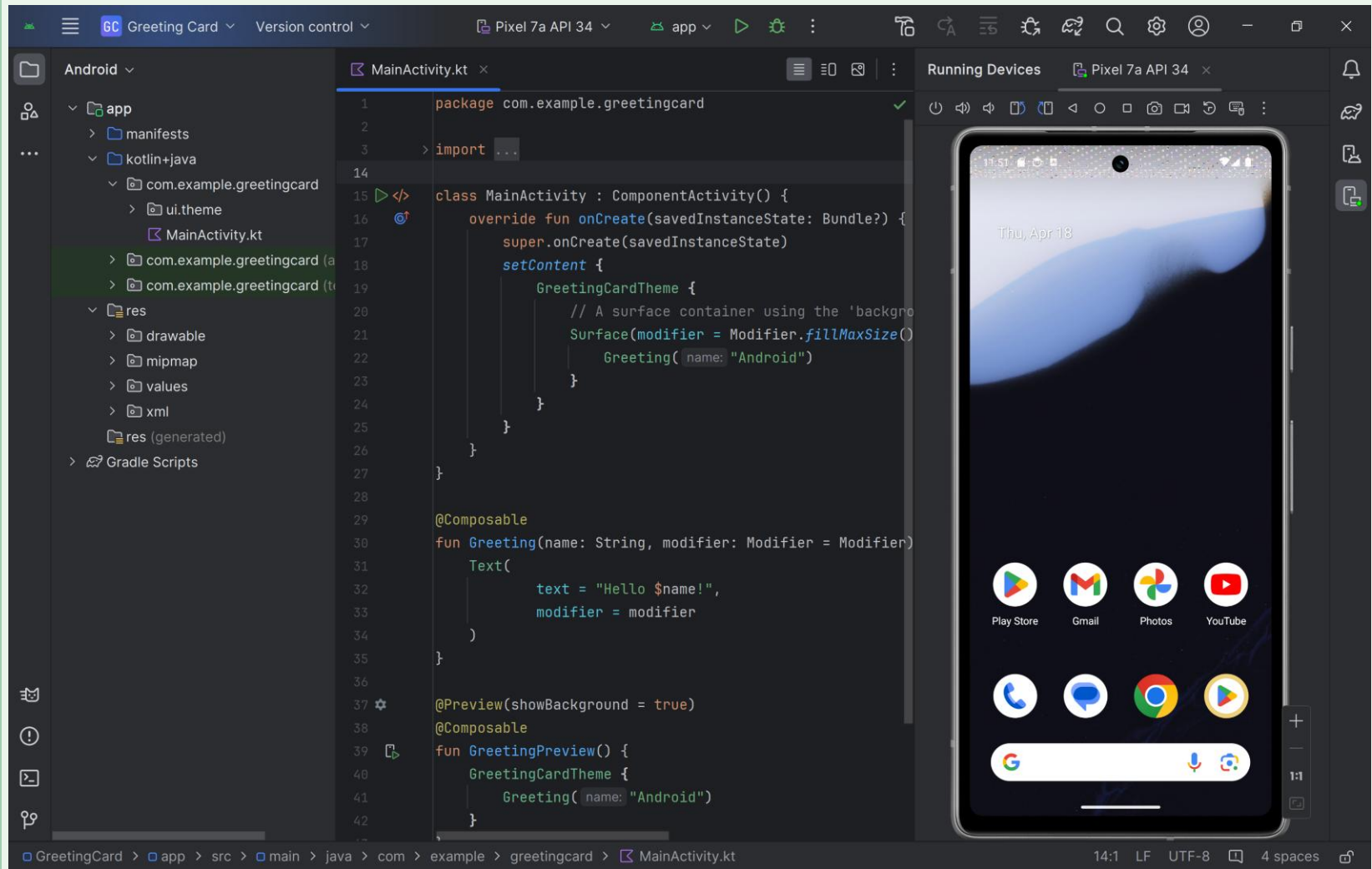


Device Manager Window





Android Studio – Project Structure





Main Activity

```
class MainActivity : ComponentActivity() {  
    override fun onCreate(savedInstanceState:  
        Bundle?) {  
        super.onCreate(savedInstanceState)  
        setContent {  
            GreetingCardTheme {  
                // A surface container using the  
                'background' color from the theme  
                Surface(modifier =  
                    Modifier.fillMaxSize(), color =  
                    MaterialTheme.colorScheme.background  
                ) {  
                    Greeting("Android")  
                }  
            }  
        }  
    }  
}
```

```
@Composable  
fun Greeting(name: String, modifier:  
    Modifier = Modifier) {  
    Text(  
        text = "Hello $name!",  
        modifier = modifier  
    )  
}  
@Preview(showBackground = true)  
@Composable  
fun GreetingPreview() {  
    GreetingCardTheme {  
        Greeting("Android")  
    }  
}
```



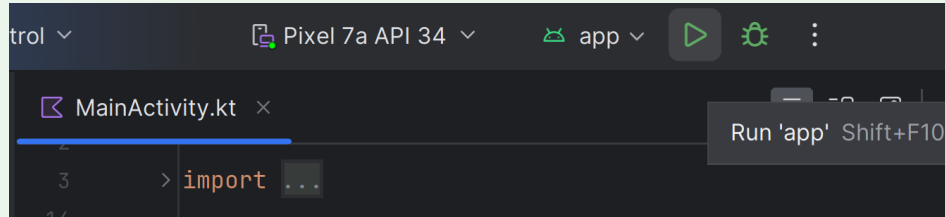
Main Activity

- ❑ Notice that the class is based on the **ComponentActivity** class.
 - An Activity is a single application entity that is used to perform actions.
- ❑ An application may have many separate activities, but the user interacts with them **one at a time**.
- ❑ The **onCreate()** method will be called by the Android system when your Activity starts — it is where you should perform all **initialization** and **UI setup**.
- ❑ An activity is not required to have a user interface, but usually will.

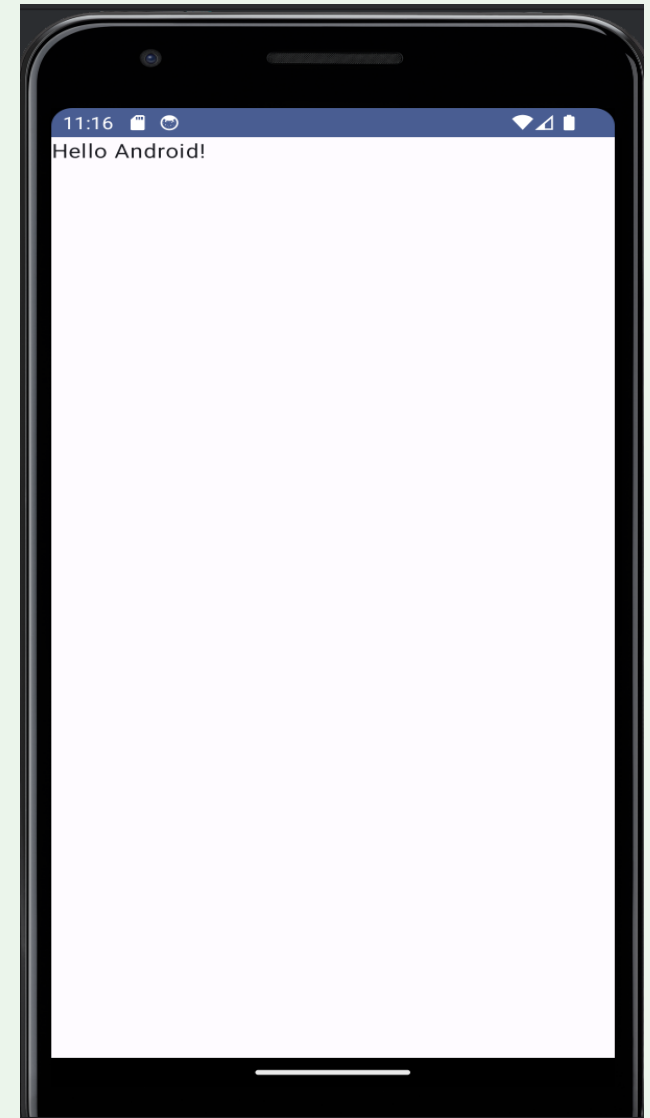


Running the application

❑ Click on the Run app icon:



❑ Wait for the emulator to load and run the app.





Developing Android Applications

- ❑ Running or debugging your application does the following behind the scenes:
 - **Compiles** the current project source **to bytecode**, and **converts** that to an Android executable (**.dex**)
 - **Packages** the executable and your project's resources and manifest into an Android package (**.apk**)
 - **Starts the virtual device** (if you've targeted one and it's not already running)
 - **Deploys your APK** to the target device and installs it
 - **Starts your application**



Creating Your First App

- ❑ **Follow the instructions here:**
- ❑ <https://developer.android.com/codelabs/basic-android-kotlin-compose-first-app#0>



References

- ❑ Textbook
- ❑ Android Documentation
 - <https://developer.android.com/guide/components/fundamentals.html>
 - <http://developer.android.com/tools/studio/index.html>
 - <https://developer.android.com/training/basics/firstapp/creating-project.html>
- ❑ Google Tutorial: <https://developer.android.com/training/basics/firstapp/creating-project.html>
- ❑ https://developer.android.com/studio/run/managing-avds?_gl=1*1y0vz1h*_up*MQ..*_ga*MTUyNDIzOTQxNy4xNzEzNDUzOTA4*_ga_6HH9YJMN9M*MTcxMzQ1MzkwNy4xLjAuMTcxMzQ1MzkwOC4wLjAuMA.
- ❑ <https://www.javaworld.com/article/3272244/what-is-the-jvm-introducing-the-java-virtual-machine.html>
- ❑ <https://www.javaworld.com/article/3304858/what-is-the-jre-introduction-to-the-java-runtime-environment.html>